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Review article

Satellite remote sensing for disaster management support: A holistic and staged approach based on case studies in Sentinel Asia

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ABSTRACT

Satellite remote sensing is one of the primary support tools for disaster management. The Sentinel Asia (SA) initiative was established in 2006 as a collaboration between regional space agencies and disaster management agencies, applying space technology (including representative satellite remote sensing) and Web-GIS technology to assist in disaster management of the Asia–Pacific region. The Japan Aerospace Exploration Agency (JAXA) worked to establish the SA framework and determine the implementation plan as an SA secretariat. SA can be regarded as an empirical research project to study how satellite remote sensing can support disaster management, in collaboration with users. This paper derives requirements for applying satellite remote sensing to disaster management support via a holistic (including human factors) and staged approach based on case studies in SA.

1. Introduction

Remote sensing [1–4] is a technology for remotely studying the properties of objects by using electromagnetic radiation, without touching the objects directly. Satellite remote sensing covers wide-ranging areas, operates continually during all hours and in all types of weather, and is used to survey Earth's surface and atmosphere to study global environmental problems, monitor disasters, explore resources, and so on. A satellite remote sensing system [1] consists of five components, as shown in Fig. 1: sources of radiation (the Sun, the Earth, and an artificial radiation source), interaction with the atmosphere, interaction with the Earth's surface, space segment (sensors), and ground segment. It should be noted that human factors (such as system operators and system users working in disaster management and response) in the ground segment as well as technical factors are important in applying satellite remote sensing to disaster management support.

The application of satellite remote sensing to disaster management support under an international framework began around 2000, when satellite-based Earth observations for rapidly assessing disaster situations globally (namely, response support) were started by the International Charter Space and Major Disasters[1] [5] (hereinafter referred to as "the International Charter") and the United Nations (UN) [6]. In Japan, the Japan Aerospace Exploration Agency (JAXA) has begun engaging in disaster management support activities using the Advanced Land Observing Satellite (ALOS) launched in January 2006. JAXA has been striving to highlight the role of Earth observation satellites in disaster management, within Japan and in the international activities of Sentinel Asia[2] (SA) and the International Charter.

According to statistics (Natural Disasters Data Book 2016 [7]), Asia has been seriously damaged by natural disasters over the last 30 years, as shown in Fig. 2. This is compounded by its high levels of population (more than half of the world's population). Disasters occurring in Asia comprise 39% of the worldwide total. The region suffers 61% of global fatalities and has 88% of the total victims associated with such disasters. In view of these circumstances, the Asia-Pacific Regional Space Agency Forum[3] (APRSAF) proposed SA in 2005 to showcase the value and impact of Earth observation technologies, combined with near real-time Internet dissemination methods and Web-GIS mapping tools for disaster management support in the Asia-Pacific region.

In February 2006, SA[2] [8–11] was established as a voluntary and best-effort basis initiative under the APRSAF and collaboration among the space community (space agencies); the international community, that is, the United Nations Economic and Social Commission for Asia and the Pacific (UNESCAP), the United Nations Office for Outer Space Affairs (UNOOSA), Association of South-East Asian Nations (ASEAN), the Asian Institute of Technology (AIT), etc.; academia (university,

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[1] The International Charter website. https://disasterscharter.org/ [accessed on September 15, 2018; the same hereinafter]
[2] Sentinel Asia website. https://sentinel.tksc.jaxa.jp/.
[3] APRSAF website. http://www.aprsaf.org/.

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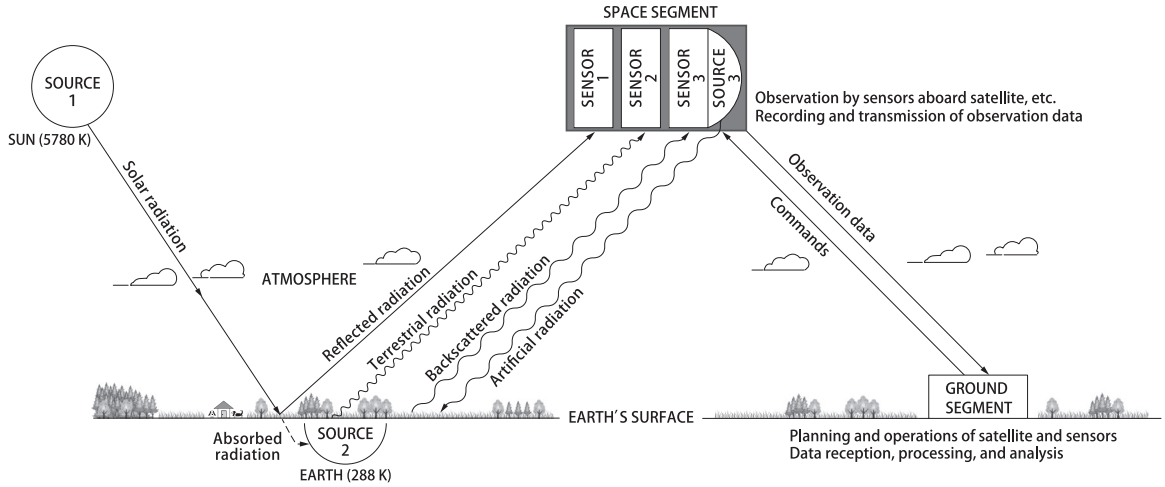


Fig. 1. Satellite remote sensing system with five components: sources of radiation (the Sun (Source 1), the Earth (Source 2), and an artificial radiation source (Source 3)), interaction with the atmosphere, interaction with the Earth's surface, space segment (sensors), and ground segment.
Source: P. J. Curran (1985) [1] (with modifications).

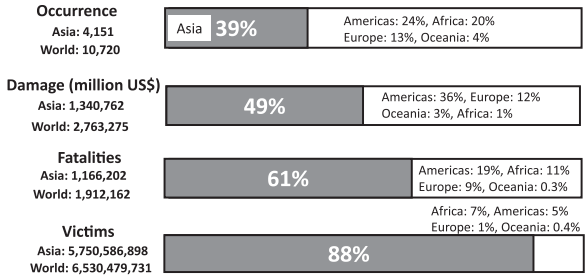


Fig. 2. Impacts of natural disasters by region, 1987–2016.
Source: ADRC-Natural Disasters Data Book 2016 [7].

research, and technical institutes); and the disaster management community (ADRC and its member organizations[4]), as shown in Fig. 3. To implement the SA, a joint project team[5] (JPT) was organized. Membership in the JPT is open to all disaster management organizations, regional/international organizations and universities/institutes that are prepared: (i) to provide their experiences, research results, and technical capabilities or (ii) to employ disaster information/data as users. JAXA worked to establish the SA framework and determine the implementation plan, as an SA secretariat.

The purpose of this paper is to study empirically how satellite remote sensing can support disaster management based on case studies in SA and to derive requirements for applying satellite remote sensing to disaster management support, by a holistic (including human factors) and staged approach.

2. Methodology

The following approaches were adopted to examine how satellite remote sensing can support disaster management and to derive requirements for applying satellite remote sensing to disaster management support, as shown in Fig. 4.

(1) Case study
JAXA has initiated case studies to examine how satellite remote sensing can support disaster management collaborating with users/end-users considering a regional background such as a disaster type, a disaster management system, and an organizational structure of disaster management, in different countries. In this paper, "case studies" refers to activities of SA to collaborate with users/end-users in different countries to demonstrate the employment of SA; "end-users" refers to local governmental/regional organizations and others working for rescue/relief/evacuation; and "users" covers a general user base including

(2) Staged approach
A 3-step approach for the implementation of SA was adopted to facilitate progressively building and expanding the regional network of data providers, data analysts, and users.

(3) Holistic approach
Based on case studies, discussions are made from the holistic viewpoint of progression level of case studies, progress of technology, user requirements, enhancement of support system, and human factors, and requirements for applying satellite remote sensing to disaster management support are derived.

3. Case studies

Case studies have been initiated from Step 2, for the employment of SA under collaboration with users (including end-users) in different countries.

3.1. Site survey after disaster response support

Disaster response support [5,6,12,13] (i.e., to observe the situation right after an extensive disaster using Earth-observation satellites to support disaster relief activity) has been the main activity of SA, which is the most suitable field for satellite remote sensing, and is greatly expected from users because satellite observation covers wide-ranging areas and operates continually, during all hours and in all types of weather.

In the framework of Sentinel Asia, JAXA conducted site surveys after disaster response support, by visiting users and end-users who worked in disaster response, to confirm in detail how satellite-based disaster data/information were employed and to discuss issues related to SA's support. For example, the following cases were conducted:

(1) Large-scale flood in Nepal in August 2008
A large-scale flood occurred after a dike burst in Sunsari district in southeastern Nepal on August 18, 2008. The changed flow of river water (from the Saptakoshi River), which entered settlements at Kushaha, Haripur, Sripur, and Laukahi village development

[4] ADRC members. http://www.adrc.asia/disaster/index.php.
[5] Sentinel Asia JPT members. https://sentinel.tksc.jaxa.jp/sentinel2/MB_HTML/JPTMember/JPTMember.htm.

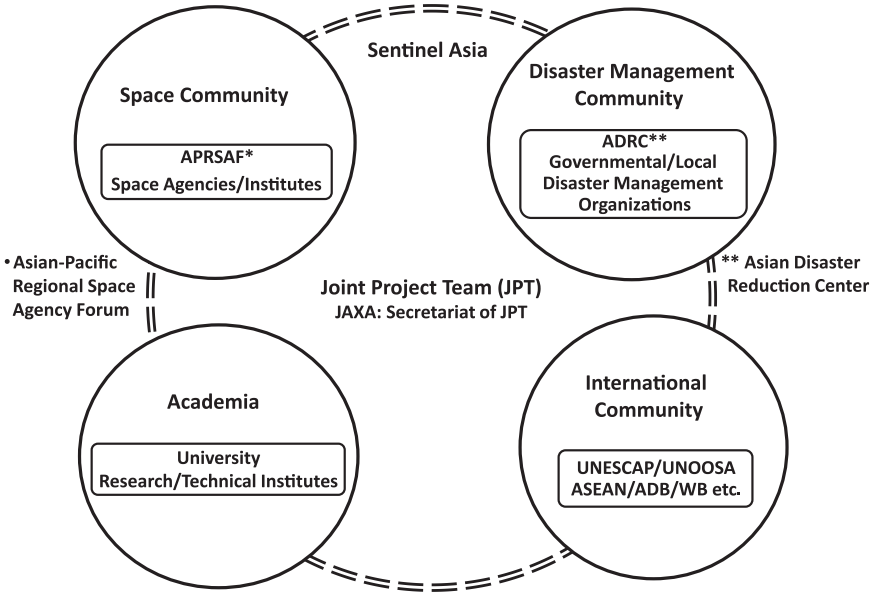


Fig. 3. Framework of SA.

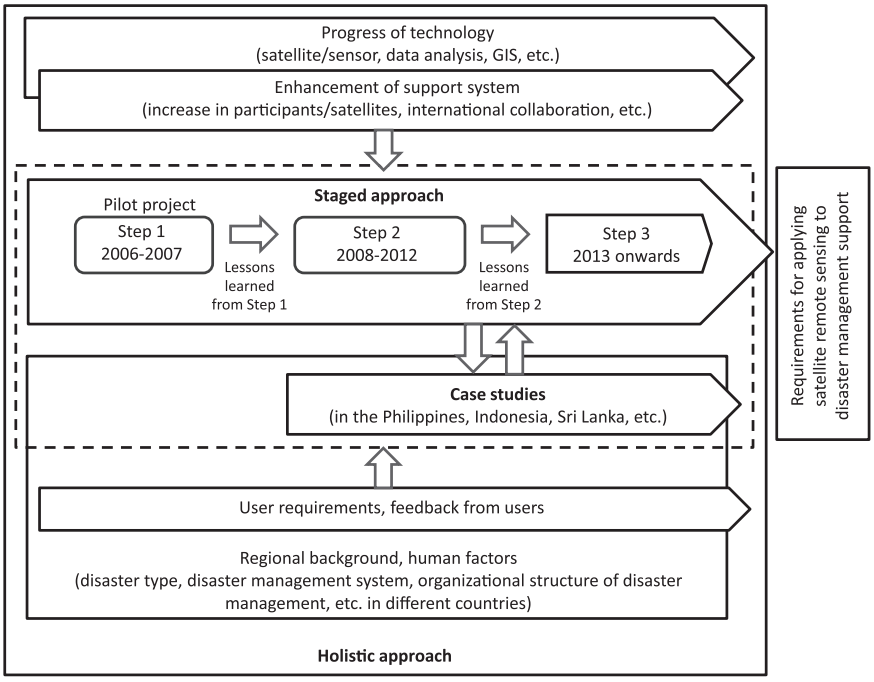


Fig. 4. A holistic and staged approach to study how satellite remote sensing can support disaster management and to derive requirements for applying satellite remote sensing to disaster management support, based on case studies in SA.

committees (VDCs) following the embankment breach, also damaged a 5-km stretch of the East–West Highway and a bridge, thereby obstructing traffic. As many as 40 thousand locals were affected, and several were missing.

JAXA made an emergency observation with the Panchromatic Remote-sensing Instrument for Stereo Mapping (PRISM) and Advanced Visible and Near Infrared Radiometer type-2 (AVNIR-2) aboard ALOS on August 22 and by PALSAR (L-band SAR) on August 24, at the SA-mediated request of the International Centre for Integrated Mountain Development (ICIMOD), Survey Department, and Department of Water Induced Disaster Prevention.

JAXA conducted a site survey in December 2008 (refer to Fig. 5(b)), and the following activities in Nepal using satellite imagery provided by SA were confirmed. Analyzed products indicating the inundation area were provided immediately to the Nepalese government. Using these data sets, overlaid with census data about population and houses, it was possible to create a map of the affected area (refer to Fig. 5(a)), and data regarding the size of the damaged area, the number of victims, and the number of houses affected could be used to rescue victims, create a recovery plan, and even manage relief payments to victims.

(2) 50-year long-term deluge in Thailand in October 2010

On the long-term deluge in October 2010 in the central and northeastern parts of Thailand, which occurs once every 50 years, JAXA made emergency observations with AVNIR-2 and PRISM aboard ALOS on October 21, 2010, at the request of the Geo-Informatics and Space Technology Development Agency (GISTDA) through SA and provided this data to GISTDA using WINDS satellite

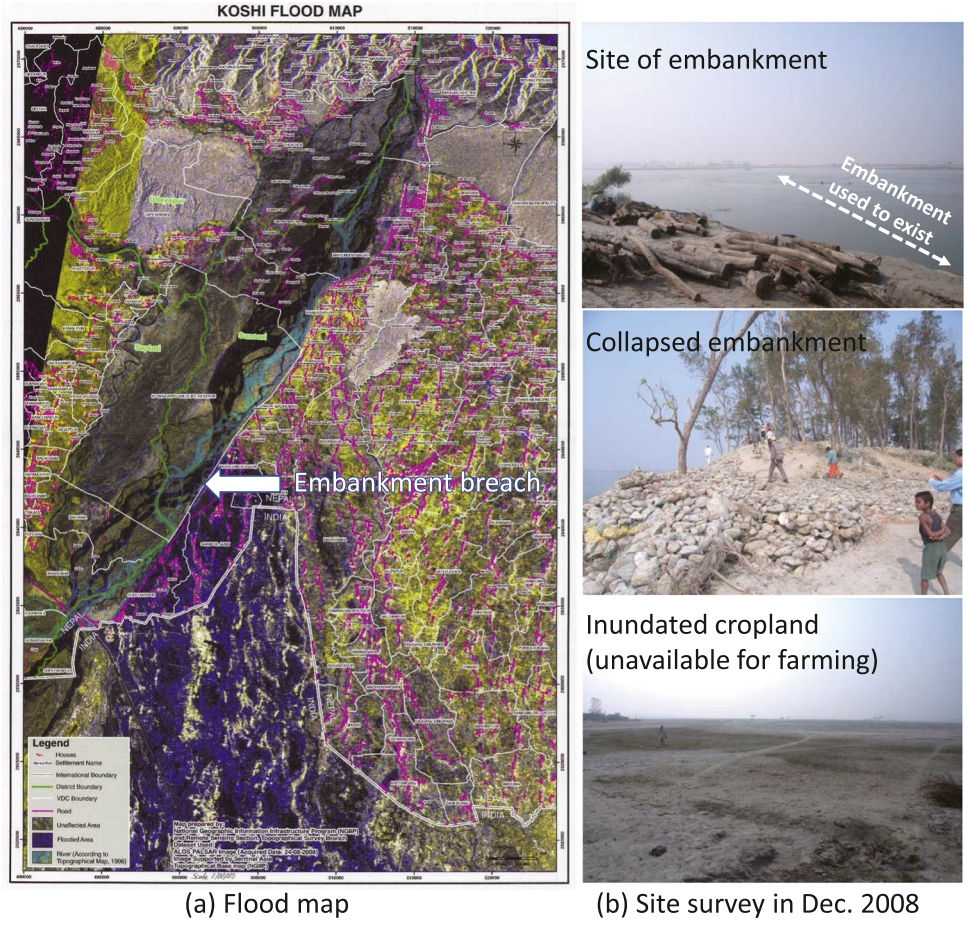


Fig. 5. Large-scale flood in Nepal in August 2008. (a) Map of the damaged area overlaid with census data, made by the Survey Department of Nepal using ALOS emergency observation imagery. This map is provided courtesy of the Survey Department of Nepal. (b) Photos were taken by ADRC in December 2008.

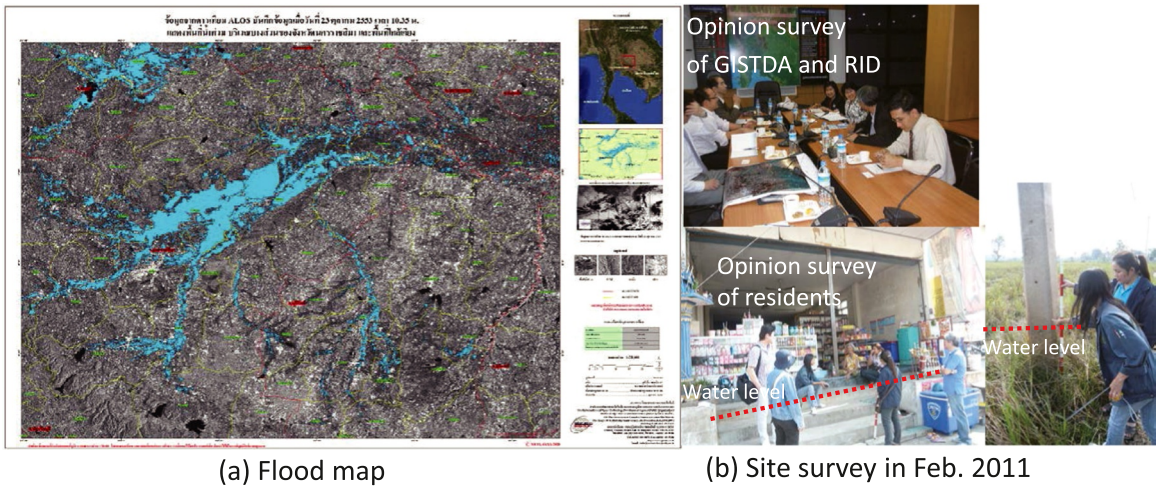


Fig. 6. Deluge in Thailand in October 2010. (a) A flood inundation map using ALOS/PALSAR imagery was made by GISTDA. The blue area shows the inundation area extracted from satellite imagery analysis. This map is provided courtesy of GISTDA. (b) An opinion survey of GISTDA and RID, and residents, as well as a field survey for validation of satellite imagery. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.).

communications.

JAXA conducted a site survey in February 2011, and made an opinion survey of governmental agencies, GISTDA and the Royal Irrigation Department (RID), and residents, as well as a field survey for validation of satellite imagery, as shown in Fig. 6(b). The following activities in Thailand using satellite imagery provided by SA were confirmed. GISTDA made inundation maps using the ALOS imagery and reported to the cabinet every day. The Thai government decided to pay compensation money to households whose homes had been directly affected. By overlaying house distribution information on a PALSAR image indicating the inundation area (refer to Fig. 6(a)), the number of suffering houses were counted and reported to the government. The Thailand government used this data to inform its relief payments.

3.2. SA Success Story in the Philippines

The SA Success Story is an activity aimed at: (1) outreach and collaboration for disaster risk reduction (DRR) in the preparedness phase, (2) establishment of case studies via regional collaboration that includes end-users, (3) local awareness and knowledge transfer through capacity building, and (4) human resources and human network development. As the first such case study, JAXA has been implementing the SA Success Story in the Philippines since 2009, responding to the needs of the Philippines for capacity building, hazard mapping using satellite imagery, a landslide early warning system using satellite-based precipitation (GSMaP[6]), and volcano and land subsidence monitoring using interferometric SAR (InSAR). Framework of SA Success Story in the Philippines is shown in Fig. 7.

3.2.1. Hazard mapping

A hazard map shows the predicted damage range by a hazard. In the framework of the SA Success Story in the Philippines, satellite imagery and data were applied to create a hazard map. By using ALOS pan-sharpened imagery and a digital surface model (DSM), hazard maps for lahars of Mt. Mayon, floods in Iloilo city, and landslides in Antique province were created by the Philippine Institute of Volcanology and Seismology (PHIVOLCS), the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA), and the Mines and Geoscience Bureau (MGB), respectively.

This mapping occurred in the first phase, from the beginning of 2009 to March 2010. Meanwhile, volcanic activity for Mt. Mayon in Luzon, the Philippines, was recorded beginning on December 14, 2009, and the lava that flowed out from the crater was confirmed on December 20. The alert level was raised to 3, prompting the order to evacuate approximately 50 thousand people living within an 8-km radius from the base of the volcano. JAXA made emergency observations with PRISM and AVNIR-2 aboard ALOS on December 25, 2009, at the request of PHIVOLCS through SA, and provided the observed data to PHIVOLCS. PHIVOLCS detected the lava deposit area using such ALOS imagery, aerial observation, and other sources, and estimated the amount of lava deposited and possible damage in the future. The lava deposit map was used in a briefing with the National Disaster Coordinating Council (NDCC) to understand the situation and make decisions. A lahar hazard map had been prepared beforehand using ALOS/PRISM DSM in cooperation between JAXA and PHIVOLCS, which was preliminarily updated by the above-mentioned lava deposit data during eruption emergencies. Later in July 2010, the Mayon volcano lahar hazard map was finally revised by PHIVOLCS as preparedness for the expected lahar hazard, as shown in Fig. 8.

3.2.2. Landslide early warning

In the framework of the SA Success Story in the Philippines, a landslide early warning system using GSMaP_NOW[6] precipitation data [14,15] has been developed under collaboration by JAXA, NTT Data, PHIVOLCS, MGB, PAGASA, and local governments in the Philippines (refer to Figs. 7 and 9), as follows:

(1) GSMaP_NOW data was corrected in real time using ground precipitation data measured by automatic weather stations (AWSs).
(2) An algorithm to determine landslides, the same one used in Japan [16], is as follows: a rainfall event is defined by short-term precipitation (hourly rainfall) and long-term precipitation (soil moisture index (SMI) calculated by tank model). The criteria of early warnings are determined as a critical line (CL) on the plane consisting of the X-axis (SMI) and Y-axis (hourly rainfall) by analyzing archived past rainfall events that did not accompany a landslide using the Radial Basis Function Network (RBFN).
(3) In 2016, a local calibration and training on the prototype system was conducted in Antipolo City and Rizal Province, aiming at the actual operation of a landslide early warning system in the Philippines.

A landslide early warning prototype system using GSMaP has been subjected to a local calibration and training program, and the decision was made to study an actual landslide early warning system covering the whole of the Philippines.

3.2.3. Volcano and land subsidence monitoring using InSAR

There are many active volcanoes in the Philippines, such as Mt. Mayon (see Fig. 8), and land subsidence in and around Manila is serious. The monitoring of volcanoes and land subsidence is an urgent task in the Philippines. SAR can measure the displacement of the Earth's surface by applying an interferometric method to two SAR images of the same area gathered from slightly separated positions, which is called interferometric SAR (InSAR) (see page 8 of reference [17]). In the framework of the SA Success Story in the Philippines, an application of InSAR technology using PALSAR data to monitor volcanic activities at Mt. Mayon, Mt. Taal, and Valley Fault and land subsidence in the Manila area has been successfully demonstrated.

3.3. Wildfire early detection and control in Indonesia

Wildfires are major and recurring phenomena that have serious impacts on property and human health, affecting many countries in the Asian region. In response to the needs of Asian countries, wildfires have been chosen as one of the SA's working group (WG) activity areas. After obtaining accurate information on the location and intensity of fires, the subsequent control of such fires is important. Furthermore, wildfires' effects are of great relevance on both a regional and global level: they bear a substantial influence on global warming, given that fires change forests, originally CO_2 sinks, into sources of CO_2 emissions.

The characteristics of the wildfire differ region by region and depend on the type of vegetation cover in the region. In general, the characteristics of wildfires in the Asia region are summarized as follows [18]:

(1) Besides ordinary forests, bushes, shrubs, and grass, there are peat bogs in Indonesia, the southern part of Thailand, and so on. Therefore, we generally call them “wildfires” not “forest fires."
(2) A large proportion of the fires are caused by humans, some of which are due to slash-and-burn farming and land clearing, and only a few of them are caused by lightning strikes.
(3) Irrigation of the peat bog and the slash-and-burn farming method are essential for farming in this region. This region is affected by wildfires and their associated smoke in the dry season and affected by floods in the rainy season every year. Floods wash away the fertile soil obtained by slash-and-burning. Therefore, slash-and-burning is required every year for farming.
(4) Peat bogs show significant characteristics from the viewpoint of carbon dioxide (CO_2) cycle. The bogs emit CO_2 not only from the peat fires but also due to activities of microbes as they become dry. It is reported that the gross volume of CO_2 emitted by peat bogs in Indonesia—2 Giga tons per year, consisting of 1.4 Giga tons due to microbes and 0.6 Giga tons due to fire—is more than the total gross volume of CO_2 emitted in Japan, 1.3 Giga tons per year [19].

In view of these circumstances, wildfire management is an urgent task in the mitigation of global warming. The Wildfire WG has been organized and operated under international cooperation in the framework of SA, based on the experience and knowledge drawn from similar

[6] JAXA/EORC website: JAXA global rainfall watch and real-time rainfall watch. Near real-time version (GSMaP_NRT): http://sharaku.eorc.jaxa.jp/GSMaP/ Quasi real-time version (GSMaP_NOW): http://sharaku.eorc.jaxa.jp/GSMaP_NOW/.

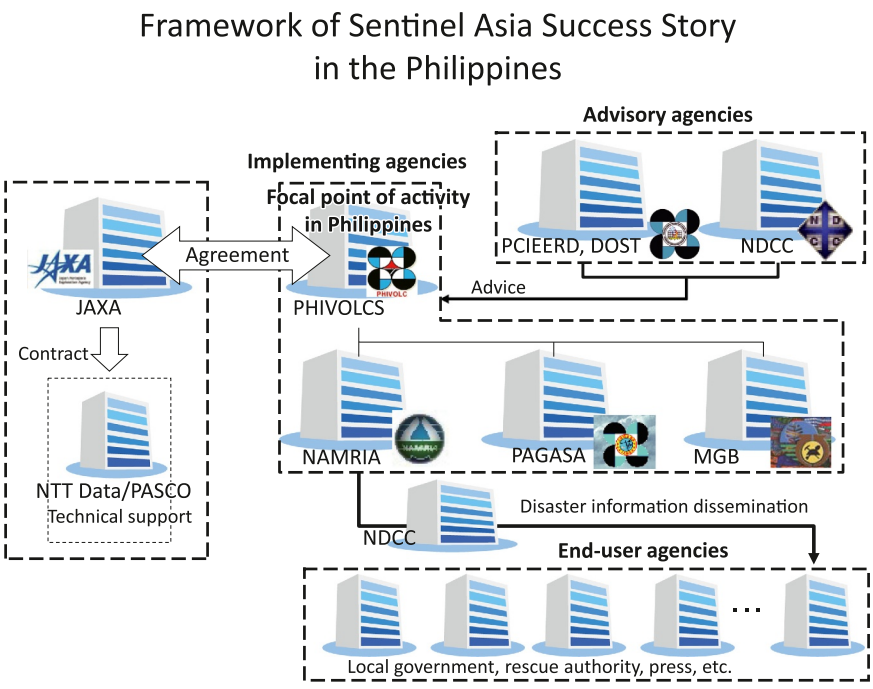


Fig. 7. Framework of the SA Success Story in the Philippines, including end-users. Notes: Philippine Institute of Volcanology and Seismology (PHIVOLCS), Philippine Council for Industry, Energy and Emerging Technology Research and Development/Department of Science and Technology (PCIEERD/DOST), National Disaster Coordinating Council (NDCC), National Mapping and Resource Information Authority (NAMRIA), Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA), Mines and Geoscience Bureau (MGB).

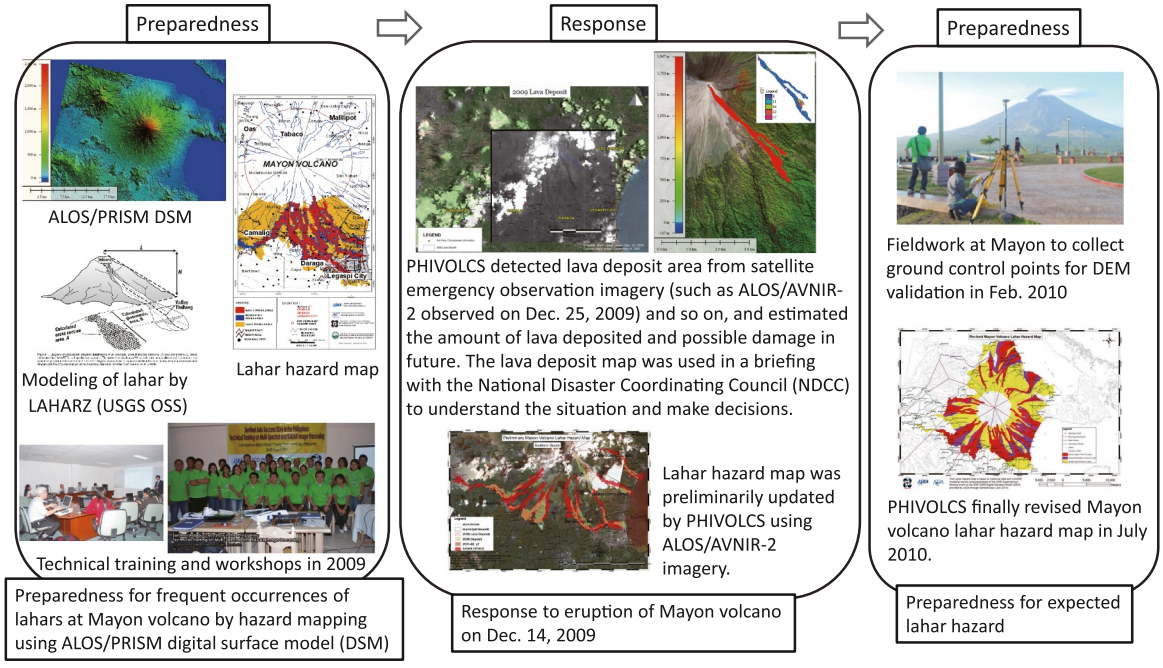


Fig. 8. Hazard mapping for lahars at Mayon volcano in the Philippines and application to response.
Source: Presentations by A. S. Daag and R. U. Solidum, Jr. of PHIVOLCS at JPT annual meetings and reply to feedback questionnaire of SA emergency observation.

activities, such as those targeting boreal-forest/tundra fires in Siberia and Alaska, bush/shrub/grass fires in Australia[7] [20], and scrub/deciduous/mixed-grass/dry-dipterocarp fires in Thailand. The goal of the SA Wildfire WG is to establish an operational cycle that includes end-users, such as firefighters, in each country to mitigate the damages caused by the fires, based on users' requirements as follows: hotspot information, which is provided by various organizations on their websites, is not reliable enough to apply it to actual fire control activities because it is difficult to distinguish critical hotspots among many hotspots on the websites. On the other hand, it is also recognized by users that satellites are useful for fire detection in the vast area. Users request SA to provide reliable and significant hotspot information, that is, to detect hotspots as precisely as possible and to distinguish critical hotspots among detected hotspots.

A typical way of effectively and comprehensively detecting active wildfires by satellites is to identify pixels that contain actively burning fires by detecting thermal infrared radiation emitted [21–23], although optical sensors can detect some wildfires by identifying their smoke. A pixel detected by the detection algorithm is called a "hotspot." A hotspot is a candidate of wildfires, because objects such as power plants and active volcanoes can also be detected as hotspots.

[7] Sentinel hotspots fire location supporting wildfire management in Australia. https://sentinel.ga.gov.au/.

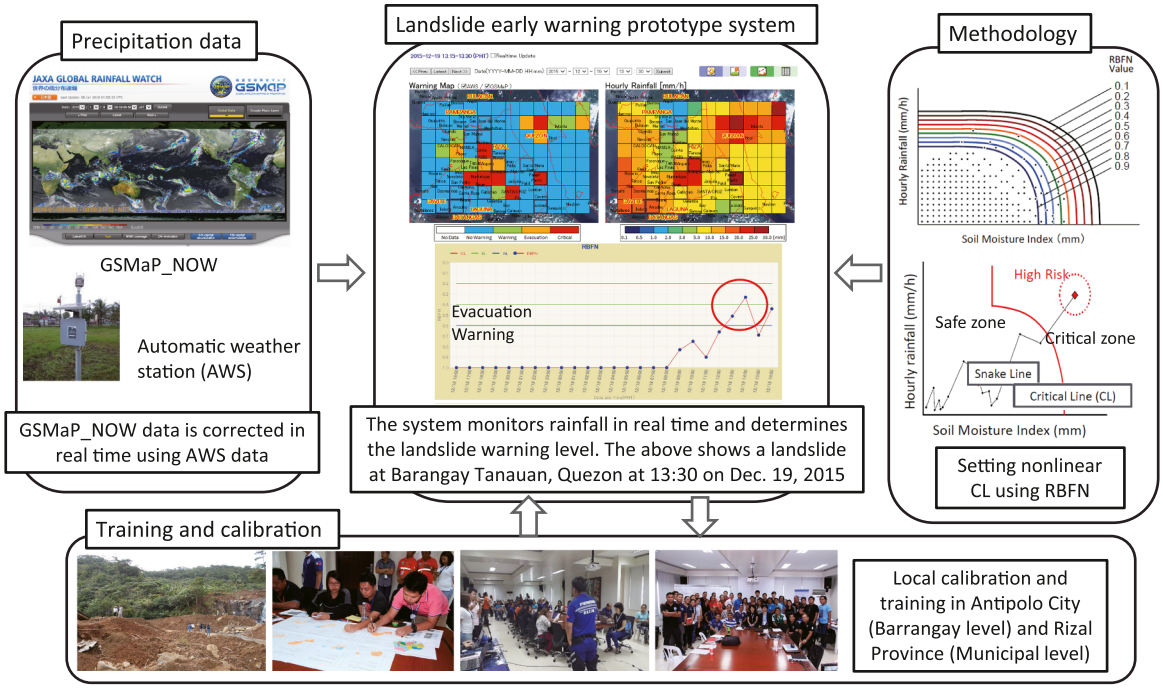


Fig. 9. Landslide early warning prototype system in the framework of the SA Success Story in the Philippines.
Source: Presentation by A. S. Daag of PHIVOLCS at APRSAF-23 annual meeting.

A demonstration project for wildfire and carbon management in a peatland in Kalimantan, Indonesia, was implemented from 2008 to 2014 under the framework of a Science and Technology Research Partnership for Sustainable Development (SATREPS[8]) by the Japan Science and Technology Agency (JST) and the Japan International Cooperation Agency (JICA). This area (Kalimantan) is characterized by being covered by a peat bog, and the interdisciplinary project primarily consists of early fire detection and control, forest (including peatland) management and conservation, and enlightenment and capacity building. The SA Wildfire WG demonstrated early fire detection and control, aiming at the establishment of an operational cycle cooperating with firefighters in Kalimantan, Indonesia [24], as shown in Fig. 10. In the demonstration, fire information (fire location and the degree of risk) was sent to firefighters organized by each villager (four villages participated in the demonstration) via short message service (SMS), where the location was expressed by a direction and distance from major landmarks such as villages and roads, according to the request of the firefighters. The project estimated the amount of CO_2 accumulation and release for tropical peatlands, proposed a prevention/early detection/fire spread prediction system of peat fires considering ground water level as one of key parameters affecting fire spread, and showed a method of carbon management and comprehensive peat land management. Due to the outcome of this project, categories of tropical peatlands are expected to be accepted by REDD-plus and others.

3.4. Glacial lake outburst flood (GLOF) early warning in Bhutan

The risk of glacial lake outburst floods (GLOFs) in the Himalayan region has been increasing due to climate change in recent years. In Bhutan, a GLOF occurred along the Po River in 1984, and five glacial lakes in the upstream regions of the Mo River were identified as being at risk of GLOFs, but their risk level was low and constant monitoring in those areas was difficult. Furthermore, due to a lack of data, it was difficult to estimate the river water levels resulting from GLOFs.

Responding to the needs of Bhutan, ADRC implemented a pilot project in Bhutan[9] from 2010 to 2012, aiming to help all ADRC member countries facing similar risks via activities such as creating hazard maps, supporting capacity building and disaster management through workshops for governments and communities, and preparing manuals related to the above-mentioned activities, based on community collaboration in the Mo River basin, in cooperation with the Department of Disaster Management (DDM), Bhutan. After creating a map with elevation data in the upstream area of the Mo River and a hazard map using ALOS imagery based on past flood records in Punakha, community-based river-level gauges were installed for a GLOF early warning system. In addition, capacity development on a community level for disaster education and training among residents was conducted. In this system, a villager living in an upstream safe area and connected to the river-level gauging system is alerted by an alarm and warns residents in downstream hazardous areas.

3.5. SA mini-project in Sri Lanka, Philippines, Bangladesh, Myanmar, and others

The SA mini-project is an activity to make disaster response support by emergency observation useful for end-users in each country. This is accomplished by (before disaster happens): (1) organizing the local community including end-users in each country, and discussing how to make high quality value-added products using satellite imagery obtained by emergency observation for end-users; (2) sharing existing maps, including hazard maps, risk maps, and ground GIS data, among local community members; (3) making new maps using Open Street Map; when disaster happens, (4) making value-added products using satellite imagery by emergency observation under collaboration with local community members as soon as possible; and sometime later, (5) conducting post-disaster evaluation of the products by site survey to employ the products for recovery and preparedness of future disasters,

[8] SATREPS: Wild Fire and Carbon Management in Peat-forest in Indonesia. https://www.jst.go.jp/global/english/kadai/h2004_indonesia.html.

[9] Glacial lake outburst floods (GLOFs) project in Bhutan by ADRC. http://www.adrc.asia/adrcreport_e/archives/2010/08/17164000.html http://www.adrc.asia/highlights/NewsNo229.

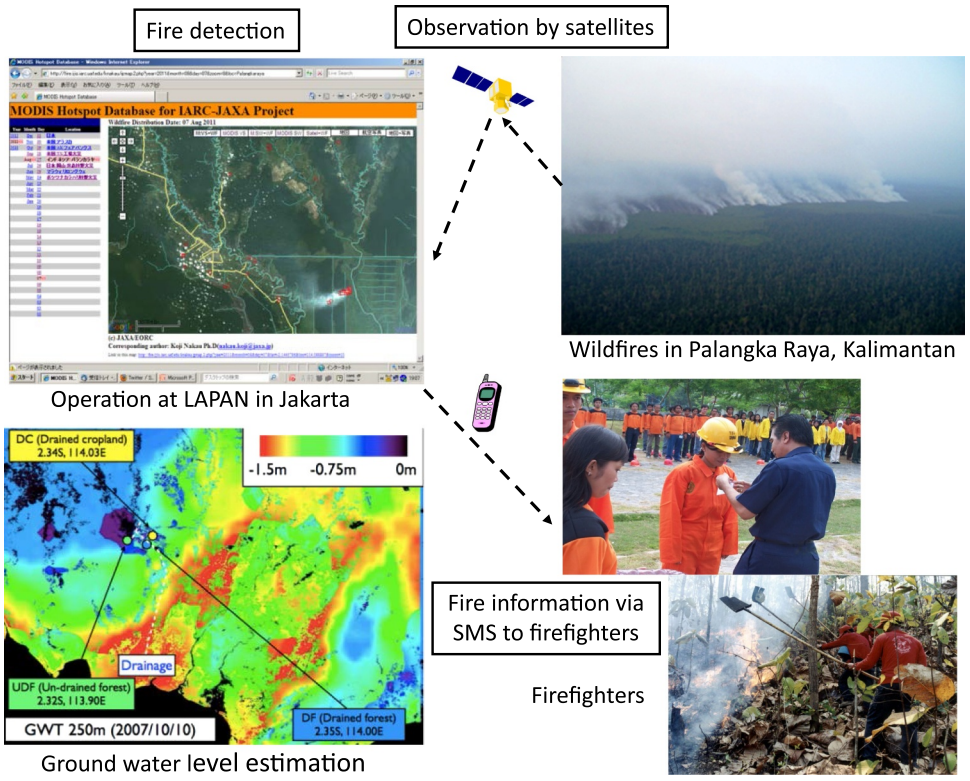


Fig. 10. Fire detection and control system of the JST/JICA project on wildfire and carbon management in a peat forest in Indonesia. The photos in the upper right and middle right are provided courtesy of A. Usup of Palangka Raya University; the lower right, the National Park, Wildlife and Plant Conservation Department, Thailand. The fire detection system in the upper left was developed by K. Nakau of Hokkaido University (then); the ground water level estimation system in the lower left, W. Takeuchi of University of Tokyo.

as well as for research purposes.

The mini-project was conducted from 2013 to 2015 in Sri Lanka, Philippines, Bangladesh, Myanmar, and so on. Through the mini-project, it was found that the scheme of making and employing value-added products in each country was not well coordinated and was not clearly defined among the related agencies enough to work in actual disaster events. In response to this, a new activity to improve the mini-project, focusing on not only preparedness but also response, has been initiated in the Philippines by the Asian Development Bank (ADB), AIT, JAXA, and related agencies in the Philippines under the support of the ADB technical assistance (TA). It aims to improve the efficiency and effectiveness of SA emergency observations and subsequent production of value-added products for disaster response support in the Philippines by: (1) optimizing SA data sharing and utilization among the related agencies with the involvement of users, (2) enhancing the capacity to process and interpret SA value-added products, and (3) making concrete work procedures and assignments of the related agencies in actual disaster events. This is expected to be expanded into other countries such as Vietnam, Indonesia, and Sri Lanka.

3.6. Response to the 2011 Great East Japan Earthquake

Among the case studies conducted by SA, the 2011 Great East Japan Earthquake was studied in the most detail (see reference [12] for details). The main lessons learned are as follows:

(1) The effectiveness of a space-based response
Although space technology is a new tool in the field of disaster management in Japan, it has been proven to effectively support relief efforts during mass catastrophes. When considering the quantity, contents, and use of satellite data by collaborative users (such as national government ministries, agencies, and local governments with which JAXA built relationships prior to the earthquake), in addition to the timing of distribution, a space-based response effectively contributes to response efforts during far-reaching catastrophes. It should be emphasized that JAXA did not accomplish this space-based response alone, but rather through partnerships with international agencies and collaborative users.

(2) The importance of international collaboration
It has also been shown that global collaboration, such as with Sentinel Asia and the International Charter, is critical and very useful for dealing with disasters. In a framework of international collaboration including the International Charter and Sentinel Asia, 14 countries and regions carried out intensive observation using 27 satellites. They helped increase the frequency of observation and approximately 5700 satellite images were provided. Global partnerships allowed for the support of various types of Earth observation satellites with respect to high-resolution and wide-ranging surveillance. In addition to measuring the general scope of the destruction by satellites with wide-ranging optics and SAR, satellites with very high-resolution optics, which were operated by the International Charter, met users' concerns about confirming the availability of roads and key facilities; this knowledge can ensure routes for evacuation, rescue, and support. It will be critical to promote future collaboration among these initiatives and establish a mechanism for working together in each country when calamities strike.

(3) The necessity of promoting awareness and support efforts to users
In the case of the Great East Japan Earthquake, collaborative users employed satellite data. The opinion survey shows that 13 (out of 28) general users employed ALOS images; most of them (12) used images provided by Iwate University which was a regional base in the Tohoku region in collaboration with JAXA. It has been shown that ALOS images were employed by users with a direct relationship to JAXA (namely, collaborative users) or an indirect relationship to JAXA (namely, via Iwate University). To have satellite images employed in disaster countermeasures by users, JAXA needs continuously promoted awareness of satellite data and support efforts to users in both ordinary times and an emergency.

(4) Promoting awareness to municipal institutions and establishing a framework
The opinion survey shows that municipal institutions used aerial

photos more frequently than satellite images; at the time, most of them were not aware of ALOS, ALOS images, and JAXA's activities. In ordinary times, it is vital to promote awareness among users, such as municipal institutions, of using satellite data for disaster management.

The opinion survey also shows that numerous general users request the establishment of a framework to supply satellite images free of charge and quickly in an emergency. To that end, in collaboration with JAXA, it will be effective to establish local bases to help impacted local governments and municipal institutions; one example is Iwate University in the case of the Great East Japan Earthquake. With respect to free of charge satellite images, all the earth observation data by satellites are greatly desired to be provided at no cost for disaster management, not only in the disaster response phase but also in the mitigation and preparedness phase and recovery phase from the higher standpoint.

(5) Wide-ranging observation and high-resolution observation

Right after an extensive catastrophe, users mostly request satellite observation to assess the general situation in damaged areas. Wide-ranging surveillance (for example, by ALOS) is most suitable. The opinion survey shows that users also request satellite observation to evaluate roads, buildings, and other circumstances in greater detail. Very-high-resolution observation is necessary for this purpose.

(6) Rapid response

In the case of the Great East Japan Earthquake, a successful rapid response occurred despite vast devastation; good weather and conveniently located damaged areas also helped. However, space agencies have been requested that continual efforts be made to shorten the response time as much as possible.

In general, when a catastrophe strikes, disaster management institutions make requests to space agencies to get information as quickly as possible. The opinion survey shows that numerous general users request the provision of satellite images free of charge and quickly in an emergency. A rapid response entails shortening the total turnaround time whenever possible.

The opinion survey shows requests concerning the timing of observation and providing data, including one hour after a disaster to assess the extent of the destruction, within two to three days to detect landslides, and one week later to estimate the quantity of debris.

4. Results and discussions

Based on case studies in Chapter 3, discussions are made from the holistic viewpoint of progression level of case studies, progress of technology, user requirements, enhancement of support system, and human factors, as shown in Fig. 4, where JAXA's activities (especially within Japan) refer to detailed technical or operational matters. In fact, in addition to its international activities with SA and the International Charter, within Japan, JAXA has been striving to highlight the role of Earth observation satellites in disaster management in conjunction with related institutions such as national government ministries, agencies, and local governments since 2006.

4.1. Progression level of case studies

Results of the case studies in Chapter 3 are summarized in Table 1, where progression of case studies is evaluated by the following criteria (referring to the NASA Application Readiness Level[10]): actual operation level (sustained use of the application system in a decision making context), prototype level (prototype application systems integrated into the end-user's operational environment), and demonstration level (demonstration in a relevant environment). The first one (hazard mapping for lahars at Mayon volcano) in Table 1 is already in actual operation; the second one (landslide early warning system using GSMaP) is at prototype level and under study for actual operation; and the third one (volcano monitoring using InSAR) is at demonstration level and under study for actual operation. These case studies in the Philippines likely to end in success are mainly due to the strong collaboration among agencies including local disaster management organizations.

To raise a case study from demonstration level to prototype level or actual operation level, it is necessary for users to recognize that satellite-based disaster information is useful for disaster management, and sustained operation by users themselves is possible.

4.2. Progress of technology

Progress of technology is discussed from the viewpoint of key technologies and new technologies, using mainly JAXA's cases.

4.2.1. Shortening of overall response time and automated satellite data analysis

In disaster response support, a rapid response, that is, to provide disaster information obtained from satellite imagery to users (disaster responders) for relief efforts as quickly as possible, is required. After receiving a request from users, the following operations are necessary: observation planning (including a change to the existing plan if necessary and allowed), commanding a satellite, satellite surveillance and recording the observation data, transmitting data to the ground, processing and analyzing that data, and adding value to the analyzed imagery, as well as coordination with users over the process and overall management of this task. A rapid response entails shortening the total turnaround time whenever possible. The following can hasten the observation process: (1) as many satellites as possible (including satellite constellations) should participate in surveillance; (2) in addition to optical sensors, radar sensors capable of operating at night or in cloudy weather should be used; and (3) automated data processing and analysis should be employed as a quick report solution.

A review of global trends in satellite-based emergency mapping from 2000 to 2014 [6] shows that the average overall response time from mobilization to first product was ~4.5 days in 2006; this was reduced to ~2.5 days on average by 2014.

In the case of JAXA, ALOS-2 with SAR, which passes over Japan twice everyday around 00:00 (JST) and 12:00 (with a margin of about 1.5 h before and after), can be operated as follows, which is called “ALOS-2 rapid response” in this paper (see Fig. 11). An emergency observation request can be accepted up to 1 h before the command uplink. In addition, it is possible to provide a quick report via automated data analysis in about two hours, and disaster area extracting information in about 5 h after observation [17]. ALOS-2 rapid response makes it possible to use satellite-based information about damaged areas observed at night (even in bad weather) for making a survey plan to be conducted in the daytime.

An example of ALOS-2 rapid response (from the original Japanese version of reference [17]) is as follows. In the morning of September 20, 2016 (9:50 JST), a request for emergency observation was issued from Tokushima Prefecture in Japan to JAXA for grasping areas where there was a possibility of flooding in the Yoshino River and Naka River basin, and the observetion was conducted by ALOS-2 in the day pass (after 12:00 pm). On the same afternoon, JAXA submitted a quick report obtained via automated data analysis (15:22) and flooded area extracting information (18:40 and 21:20) to the Tokushima Prefecture and shared data with related departments. In the Tokushima prefecture, "Utilization of satellite information in grasping the disaster situation," cooperated with JAXA, was included in the revision of the Tokushima prefecture disaster prevention plan in October 2017.

Quick reports obtained via automated data analysis compare an

[10] NASA Application Readiness Levels (ARLs) https://www.nasa.gov/sites/default/files/files/ExpandedARLDefinitions4813.pdf.

Table 1
Achievements and progression level of case studies.

Disaster Management Phase	Case studies	Country	Used satellite data and technology	Achievements in case studies	Progression level (*1)	Notes
Preparedness	Hazard mapping for lahars at Mayon volcano (refer to Section 3.2.1)	Philippines	– ALOS/PRISM Digital Surface Model (DSM) – ALOS/AVNIR-2 – Hazard mapping	On eruption in Dec. 2009, the lava deposit map using satellite imagery etc. was used in a briefing with the National Disaster Coordinating Council (NDCC) to understand the situation and make decisions.	Actual operation level	Hazard map has been maintained and revised by the Philippine Institute of Volcanology and Seismology (PHIVOLCS).
	Landslide early warning system in the Philippines using GSMaP (Section 3.2.2)	Philippines	– GSMaP_NOW – Real-time correction of GSMaP_NOW data – Radial Basis Function Network (RBFN)	A prototype system is undergoing a local calibration and training program, with a goal of actual operation covering the whole of the Philippines.	Prototype level	Preparations are underway for actual operation in the Philippines. An algorithm to judge landslides is the same as one in actual operation in Japan.
	Volcano and land subsidence monitoring using InSAR (Section 3.2.3)	Philippines	– ALOS/PALSAR – Interferometric SAR (InSAR)	An application of InSAR technology using PALSAR data to monitor volcanic activities at Mt. Mayon, etc. has been successfully demonstrated.	Demonstration level	Actual operation and sustained use of volcano monitoring by PHIVOLCS using ALOS-2/PALSAR-2 is under study. In Japan, this has been in actual operation by the Geospatial Information Authority of Japan (GSI) since 2014. (see Section 4.2.5)
	Wildfire early detection and control in Indonesia (Section 3.3)	Indonesia	– Terra and Aqua/MODIS – Hotspots detection	Early fire detection and control were demonstrated, aiming at the establishment of an operational cycle cooperating with firefighters in Kalimantan.	Demonstration level	If the accumulation of carbon in the peatlands is recognized in the framework of REDD-plus, implementation of this system at an actual operation level may be rapidly advanced.
	Glacial lake outburst flood (GLOF) early warning in Bhutan (Section 3.4)	Bhutan	– ALOS/AVNIR-2 – Hazard mapping	A hazard map using satellite imagery based on past flood records in Punakha and community-based river-level gauges were installed for a GLOF early warning system.	– Actual operation level (education and training) – Demonstration level (GLOF early warning system)	Along with the JICA project for Capacity Development of GLOF and Rainstorm Flood Forecasting and Early Warning in Bhutan in 2013–2016, GLOF early warning and disaster education/training have been carried out by themselves.
	SA mini-project in Sri Lanka, Philippines, Bangladesh, Myanmar, etc. (Section 3.5)	Sri Lanka, Philippines, Bangladesh, Myanmar, and others	– ALOS/AVNIR-2 and PALSAR – Hazard mapping – Emergency mapping	An activity to make disaster response support by emergency observation useful for end-users in each country was conducted.	Demonstration level	The scheme was not well coordinated and defined enough to work in actual disaster events. In response to this, a new activity to improve the mini-project has been initiated in the Philippines.
Response	Large-scale flood in Nepal in August 2008 (Section 3.1(1))	Nepal	– ALOS/PALSAR – Emergency mapping	The Nepalese government used this map to rescue victims.	Demonstration level	No information on progress after this flood event
	The 2011 Great East Japan Earthquake (see Section 3.6 and reference [12])	Japan	– ALOS/AVNIR-2, PALSAR and others – Emergency mapping	Although satellite remote sensing was a new tool in the field of disaster management in Japan, it has been empirically proven to effectively support relief efforts during mass catastrophes.	Demonstration level	By March 14, 2011, cloud-free images from ALOS, etc. were obtained for nearly all coastal areas, which comprised the first post-disaster data and the most frequently used geospatial results during the early response stages.
Recovery	Large-scale flood in Nepal in August 2008 (Section 3.1(1))	Nepal	– ALOS/PALSAR – Emergency mapping	The Nepalese government used to create a recovery plan and manage relief payments to victims.	Demonstration level	No information on progress after this flood event
	50-year long-term deluge in Thailand in October 2010 (Section 3.1(2))	Thailand	– ALOS/PALSAR – Emergency mapping	The Thailand government used to inform its relief payments.	Demonstration level	No information on progress after this flood event

(*1) Progression level[10]: Actual operation level (sustained use of the application system in a decision making context), prototype level (prototype application systems integrated into the end-user's operational environment),and demonstration level (demonstration in a relevant environment)

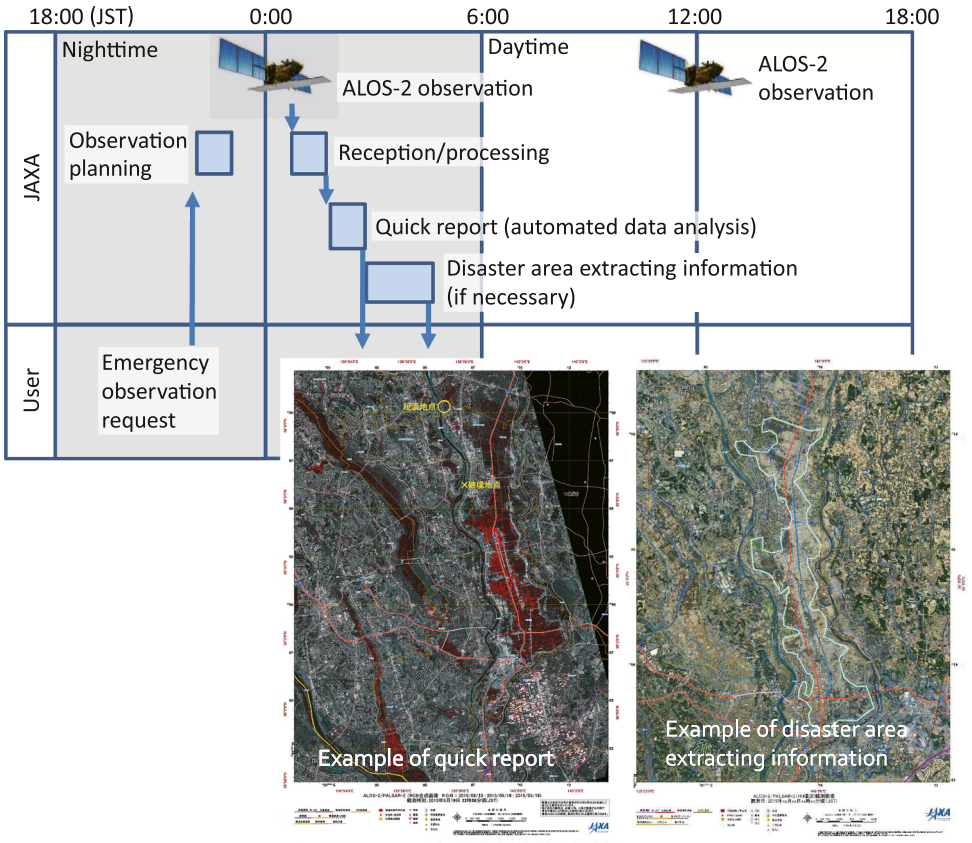


Fig. 11. ALOS-2 rapid response.

image of emergency observation with an image before the disaster and extract the changed parts. For example, if red is assigned to an image before disaster and green and blue are allocated to an image after disaster obtained by emergency observation to create a color composite image, the red part of the composite image will indicate flooded areas, because the red part of the composite image is the area displayed in red (ground with reflection) in the pre-disaster image and not colored (water surface with no reflection) in the post-disaster image.

4.2.2. *Progress of spatial resolution and swath width of sensors*

The spatial resolution of a sensor, which determines objects a sensor can identify, is defined by the size of the surface area corresponding to an individual pixel of a sensor, where a pixel is a single detector element of a sensor. By zooming in a satellite image displayed on a screen, a square mosaic appears, which shows the pixels of the image. The spatial resolution of a sensor is determined by an instantaneous field of view (IFOV) of the sensor and the distance from the object. A sensor at a low altitude has a higher spatial resolution than the same one would at a higher altitude.

The belt-shaped area on the Earth's surface imaged by a sensor with the movement of the satellite is referred to as the swath. The spatial resolution and swath width of a sensor are set for each satellite based on its mission requirements and data handling capacity. Generally, if the spatial resolution is higher, the swath width is required to be smaller, as exemplified in Fig. 12. A higher spatial resolution or wider swath requires more pixels, which causes the data volume to become larger and make data transmission and handling more difficult.

From the viewpoint of disaster monitoring, right after an extensive catastrophe, users mostly request satellite observation to assess the general situation in damaged areas, for which wide-ranging surveillance is most suitable. Users also request satellite observation to evaluate roads, key facilities, and other circumstances in greater detail to confirm their availability and ensure routes for evacuation, rescue, and support, for which very-high-resolution observation is necessary [12].

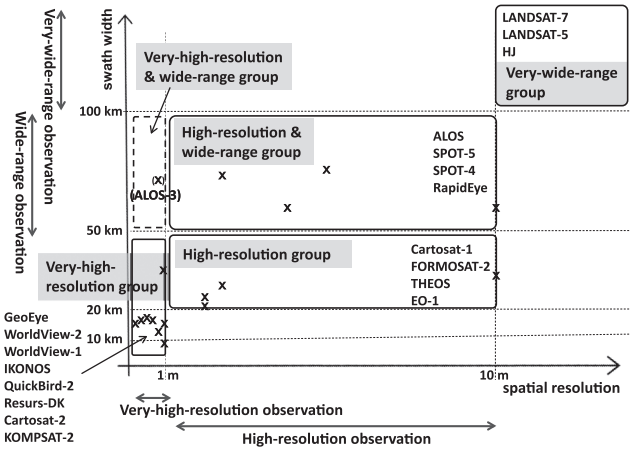


Fig. 12. A classification of optical earth observation satellites with respect to spatial resolution and swath width, which supported the response to the 2011 Great East Japan Earthquake [12]. (X) shows the Advanced Optical Satellite (ALOS-3) with 0.8 m of spatial resolution and 70 km swath under development by JAXA [25]. Note: “Very high resolution” means a spatial resolution of less than or equal to 1 m, and “high resolution” means more than 1 m and less than or equal to 10 m. “Very wide range” means a swath of more than 100 km; “wide range” signifies more than 50 km and less than or equal to 100 km. In the classifications above, the best values for resolution and related swath width for each satellite are based on information found on the Internet.

In the case of JAXA, the spatial resolution and swath width have been improved as follows. ALOS with an optical sensor (2.5 m of spatial resolution and 70 km swath) was operated from 2006 to 2011. JAXA is

planning to launch a highly advanced Earth observation satellite ALOS-3 [25] around 2020–2021 with an optical sensor that has 0.8 m of spatial resolution and 70-km swath. This satellite belongs to the very-high-resolution and wide-ranging group in Fig. 12, which is suitable for disaster monitoring. As for SAR, its high-resolution mode can observe with 7–44 m of spatial resolution and 40–70-km swath (ALOS), 3 m of spatial resolution and 50-km swath (ALOS-2 in operation since 2014), and 3 m of spatial resolution and 200-km swath (ALOS-4 to be launched around 2020–2021); while its wide-ranging mode can observe with 100 m of spatial resolution and 250–350 km swath (ALOS), 60 m of spatial resolution and 490 km swath (ALOS-2), and 25 m of spatial resolution and 700 km swath (ALOS-4).

4.2.3. Sharing/providing of data/information using Web-GIS and satellite communications

A sharing/providing system of data/information is essential for conducting disaster management support. In the case of SA, the information-sharing platform from Step 1 was constructed using the Digital Asia Web-GIS of Keio University with dispersed data at each data provider. To extract disaster-related information, it is important to have value-added products, which involve overlaying satellite images with various GIS data such as census data and general/hazard/risk maps. Thus, GIS data should be produced during the preparedness phase. A new sharing/providing platform of data/information was developed and operated by JAXA as a part of Step-2 activities. This is a Web-GIS with centralized data at JAXA in Japan. The Step-2 platform is a robust system with redundant construction and centralized data management at JAXA. It functions to share information and data on Web-GIS, and to disseminate data to regional servers via the internet and the WINDS satellite (used between 2010 and 2017). It also has a secure-access control function to limit users from viewing and downloading data according to the data policies of the data providers. Data and information sharing through this system is illustrated in Fig. 13. In addition to information sharing via the internet (still in operation), information transmission to facilitate access to disaster-related information using WINDS satellite communications was demonstrated during 2010–2017. Regional servers mirroring the central server in Japan were set up in 11 countries. Any of these servers could be accessed by users to view the SA website or to download data. In case of a disaster, large volumes of satellite imagery could be distributed, using WINDS satellite communications and the internet, to organizations handling emergency situations as well as organizations analyzing satellite imagery.

4.2.4. Satellite-based precipitation (GSMaP) and real-time correction

JAXA provides global satellite-based hourly 10-km-grid rainfall maps called GSMaP[6] [14,15] in near real-time (about four hours after observation) using the combined microwave-infrared (MW-IR) algorithm with microwave radiometers (such as GPM-Core GMI, TRMM TMI, GCOM-W AMSR2, DMSP series SSMIS, NOAA series AMSU, and MetOp series AMSU) data and geostationary meteorological satellite infrared data. Data of the quasi real-time version (GSMaP_NOW[6]) over the geostationary meteorological satellite "Himawari" area are also available since November 2015.

GSMaP is superior in terms of extensibility, immediacy, and convenience, and it is used as input to a flood simulation system for flood risk assessment in disaster management planning and raising awareness of people concerned about disaster prevention (noted by H. Baba of JICA). However, GSMaP contains errors and cannot be used as is for critical applications such as flood warnings. GSMaP includes errors caused by the principle of microwave radiometers (estimating rainfall near the Earth's surface from the amount of ice in the sky), observation resolution (10 Km grid of spatial resolution) and globalization of data by interpolation (interpolation of microwave radiometer data using a cloud movement vector calculated from infrared radiometer data of a geostationary meteorological satellite, and so on).

GSMaP is like ground precipitation data in qualitative strength of rainfall and distribution of rainfall, but it tends to be quantitatively underestimated. This suggests that GSMaP can be corrected by a linear transformation, referring to ground precipitation. The correction method developed by JAXA and NTT DATA realized the rainfall accuracy applicable to flood warning: for daily rainfall data, the degree of correlation with ground data increased from 0.5 to 0.7 (reported by K. Tsutsui of NTT DATA at the 7th GEOSS Asia-Pacific Symposium in Tokyo on May 27, 2014). In the landslide early warning prototype system in the Philippines (see Section 3.2.2), GSMaP_NOW data is corrected in real time using ground precipitation data measured by automatic weather stations (AWSs).

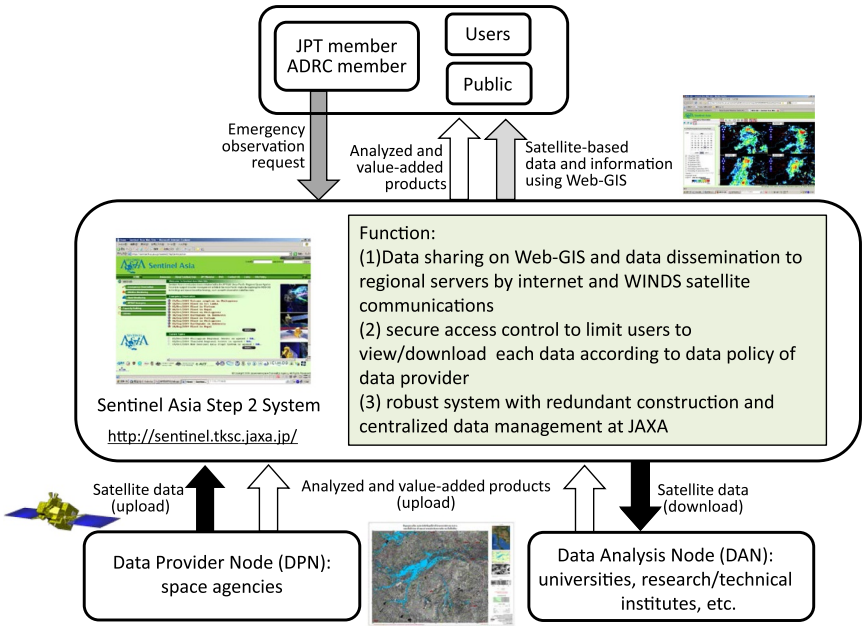


Fig. 13. SA Step-2 system (Web-GIS) for data and information sharing and dissemination.

4.2.5. InSAR for volcano monitoring

An image obtained by calculating the phase difference from two observations for each pixel on the two-dimensional SAR image is called an "interferometric SAR (InSAR) image" (see page 8 of reference [17]). It makes it possible to capture the amount of change of the ground surface from the direction of the satellite in a planar manner. Crustal deformation can be extracted on the scale of several centimeters by InSAR analysis.

The Geospatial Information Authority of Japan (GSI) has been monitoring active volcanoes in Japan since 2014, using ALOS-2 InSAR. Upon the activation of Mt. Hakone on May 4, 2015, emergency observation by ALOS-2 was carried out at the request of the Volcano Working Group (secretariat: Japan Meteorological Agency) established under the Coordinating Committee for Prediction of Volcanic Eruptions in Japan. Results of InSAR analysis by the Hot Spring Research Institute of Kanagawa Prefecture and the GSI were reported to Hakone Town and the Hakone Volcano Coordinating Committee and were used for decision making of the re-opening of restricted areas and safety measures [17].

In the framework of the SA Success Story in the Philippines, an application of InSAR technology using PALSAR data to monitor volcanic activities at Mt. Mayon, Mt. Taal, and Valley Fault and land subsidence in the Manila area has been successfully demonstrated (see Section 3.2.3). Actual operation and sustained use of volcano monitoring in the Philippines using ALOS-2 InSAR is under study.

4.3. User requirements

As shown in Fig. 4, user requirements are one of the major factors to promote and enhance SA's activities which have been collected and discussed at the JPT[5] meeting held twice or once a year, starting with the JPT meeting at Hanoi, Vietnam in February 2006 (see Section 3.1 of reference [8] for details). In addition, JAXA conducted an opinion survey for users in Japan after the 2011 Great East Japan Earthquake between August 2011 and March 2012, asking them about responses to the earthquake to clarify how satellite images were used in disaster countermeasures, and to identify factors that prevented such data from being used, which will enhance how satellite information is employed in the future. In total, 29 institutions (such as municipalities in damaged areas, universities, research organizations, fisheries, agricultural cooperatives, private corporations, and NPOs) were interviewed to clarify the purposes of satellite information, appropriate timing for providing it, the effects of using it, and related concerns (see Section 5.2 of reference [12] for details). Furthermore, JAXA carried out an opinion survey in 2012–27 regional disaster management authorities (23 organizations of which are ADRC members[4]) of 27 Asian countries to promote SA and to learn more about specific user requirements. These user requirements are cited as required in this paper.

For emergency observations, a questionnaire survey has been conducted to obtain feedback on the disaster response results from users. However, there is the problem that the collection rate of answers of this questionnaire survey is low. Feedback from users on each disaster event is important and valuable to data providers, whether it goes well or not. If not successful, user feedback contains lessons learned for improvement of activities; if successful, it is a measure and evidence for the results of data providers.

4.4. Enhancement of support system

Discussions are made from two perspectives: staged progress of SA and progress by international collaboration.

4.4.1. Staged progress of SA

SA followed a 3-step approach; Step 1 started in 2006; Step 2 in 2008; Step 3 began in 2013 and is in progress. Step 1 began with sharing satellite data on Web-GIS, which was considered useful for disaster management. During Step 2, the lessons and issues of Step 1 were examined and reflected in Step-2's activities. During this period, knowledge on how space technologies can support disaster management was acquired, and SA Step 3 has started. Case studies have been initiated from Step 2, for the employment of SA under collaboration with users (including end-users) in different countries. When SA was launched in 2006, only JAXA was providing satellite data to SA, and eight agencies are providing as of September 2018; similarly, the number of Joint Project Team (JPT) members[5] increased from 27 to 108.

Currently, for further enhancement of SA support, efforts called “SA evolution” are being performed. At APRSAF-21, held in Tokyo in December 2014, the necessity of evolving SA to activate and reinforce its expected functions was confirmed. A steering committee was established, and a short-term plan and a long-term plan have been prepared. The short-term plan consists of three concepts: (1) “strategic management” by the steering committee, (2) “joint drive” to encourage participating organizations to be actively involved, and to strengthen and make activities more effective and sustainable cooperation with development agencies, such as ADB, JICA, and the World Bank (WB), as well as with other relevant global frameworks, and (3) “end-user-oriented approach” to respond to the needs of disaster areas and communities directly and quickly by accumulating information both before and after disasters, such as hazard maps, as well as education, through which communities can enhance their capability to respond when disasters happen.

4.4.2. International collaboration

As many satellites as possible are expected to support disaster management, particularly response, to maximize results based on: (i) the frequency of observation; (ii) rapid response, that is, to provide satellite-based disaster information to users (disaster responders) as quickly as possible (see Section 4.2.1); (iii) wide area coverage; (iv) types of satellite data with respect to high-resolution or wide-ranging surveillance, as well as SAR (X- or C- or L-band) or optical sensors; and (v) the uncertainty of getting excellent imagery depending on weather conditions. In this context, international collaboration is useful and important, particularly in the case of catastrophic disasters such as the 2011 Great East Japan Earthquake [12], where some international space-based initiatives are contributing to the field of disaster management. Such initiatives include SA, the International Charter, the International Working Group on Satellite-based Emergency Mapping (IWG-SEM) by a voluntary group of organizations involved in satellite-based emergency mapping, the Committee on Earth-observation Satellites (CEOS), Copernicus (a European Union program), UN-SPIDER (a United Nations program), and the Global Earth-observation System of Systems (GEOSS) of GEO.

Formal collaboration between SA and the International Charter began in 2010, and ADRC began to also escalate SA's emergency observation requests for major disasters to the International Charter, to further increase available satellite resources to observe disaster-affected areas.

4.5. Human factors

In applying satellite remote sensing system (see Fig. 1) to disaster management support, human factors (as system operators and system users working for disaster management and response), as well as technical factors, in the ground segment are important.

As the results of Step 1, maintaining a good human network, between the space and disaster management communities, has been recognized to be the foundation and lifeline of the project, and just as important as the satellites and infrastructure that were implemented for this system.

Furthermore, the necessity of collaboration with end-users for the employment of SA was recognized in the early Step-2 period and since

then JAXA has initiated case studies to collaborate with users (including end-users) considering regional background such as disaster type, disaster management systems, and organizational structure of disaster management, in different countries.

In collaboration with users, a face-to-face human network is an important underlying elemrnt. It is also important to collaborate with the appropriate organization/section/person in each country according to the theme, but it is not easy because this depends on the organizational structure of disaster management in each country. After all, collaboration is done by people; in this sense, the most important thing is the human network.

Furthermore, users are neither existing nor given ones; they must be developed and maintained as a part of project activities, by sustained efforts of the human resource development and human network development, starting with awareness, education, training, and so on. In fact, this is indicated by the case of the 2011 Great East Japan Earthquake (refer to (3) and (4) of Section 3.6), as follows: collaborative users (such as national government ministries, agencies, and local governments with which JAXA built relationships prior to the earthquake) employed satellite data, whereas municipal institutions used aerial photos more frequently than satellite images; at the time, most of them were not aware of ALOS, ALOS images, and JAXA's activities. In ordinary times, it is vital to promote awareness among users, such as municipal institutions, of using satellite data for disaster management [12].

The goal of SA is to be much more than just the delivery of satellite Earth observation data, but rather to provide more specific “services” quickly (see Sections 4.2,1) on a 24-h, 7-day, 365 days basis. From the viewpoint of system operation, it is desirable to automate the system as much as possible: e.g., automated satellite data analysis (see Section 4.2.1) and delivery of disaster-relevant information products to users.

5. Conclusions

Through case studies, particularly the 2011 Great East Japan Earthquake [12], space-based response has been empirically proven to effectively support relief efforts during mass catastrophes (see (1) of Section 3.6). Furthermore, based on these case studies, requirements for applying satellite remote sensing to disaster management support can be derived from the holistic viewpoint including human factors, as follows (refer to Fig. 14):

(1) Framework

The opinion survey shows that numerous general users request the establishment of a framework to supply satellite images free of charge and quickly in an emergency (see (4) of Section 3.6). A framework including data providers (space agencies), data analysts (universities, research institutes, etc.) and users (disaster management organizations) is necessary. In the case of SA, it was established as a voluntary and best-effort basis initiative under the APRSAF and collaboration among the space community (space agencies), the disaster management community (ADRC and its member organizations, the international community, and academia (university, research and technical institutes), as shown in Fig. 3.

(2) Activities

Users request to cover the entire disaster management cycle (preparedness, response, and recovery). In the case of SA, for preparedness support, capacity building, hazard mapping, and an early warning system have been implemented (refer to Sections 3.2–3.5). It should be noted that the capacity building and early warning system align with the Sendai framework for disaster risk reduction (DRR) 2015–2030 [26].

Disaster response support (that is, to observe the situation right after an extensive disaster using Earth-observation satellites to support disaster relief activity) is the most suitable field for satellite remote sensing, and is greatly expected from users, because satellite observation covers wide-ranging areas and operates continually, during all hours and in all types of weather. A review of global trends in satellite-based emergency mapping from 2000 to 2014 [6] shows that satellite-based emergency mapping has been most intensively deployed in Asia and Europe, and SA activities have a strong prevalence in Asia, accompanied by the International Charter and UN mapping efforts. In future, it should be clarified how pre-disaster monitoring for impending disasters (which is a strong requirement from users) can be supported, considering the data policy of each space agency.

As for recovery support, products produced by emergency observation can be utilized in the recovery phase as well (as described in Section 3.1), and to prepare for recurring disasters (refer to Fig. 8). In future, it should be clarified how this phase can be supported, considering the data policy of each space agency and making use of open and free satellite data such as MODIS and Himawari (geostationary meteorological satellite of JMA).

Case studies and human networking through collaboration with users (including end-users), considering a regional background such as disaster type, disaster management systems, and organizational structure of disaster management in different countries are inevitable activities. It is also important to implement the theme of activities that meet the needs of users.

(3) Users

SA is characterized by collaboration with users who are working for disaster management and response. Users are neither existing nor given ones; they must be developed and maintained as a part of project activities, by sustained efforts of the human resource development and human network development, starting with awareness, education, training, and so on.

In collaboration with users, a face-to-face human network is an important underlying element. It is also important to collaborate with the appropriate organization/section/person in each country according to the theme, but this is not easy because it depends on the organizational structure of disaster management in each country. After all, collaboration is done by people; in this sense, the most important thing is the human network.

(4) Data providers (space agency and others)

Space agencies have been requested continual efforts to realize following user requirements.

The opinion survey [12] shows requests concerning the timing of observation and providing data, including one hour after a disaster to assess the extent of the destruction, within two to three days to detect landslides, and one week later to estimate the quantity of debris. The goal would be much more than just the delivery of satellite Earth observation data, but rather to provide more specific “services” quickly, on a 24-h, 7-day, 365 days basis.

Data users are also deeply concerned about confirming the availability of roads and key facilities immediately after a mass calamity to ensure routes for evacuation, rescue, and support (availability monitoring), in addition to grasping the overall extent of the damage (overall monitoring) [12]. For overall monitoring, wide-ranging optical/SAR images are used. For availability monitoring, very-high-resolution optical images are necessary. Specifically, resolution of less than 1 m is required. In the case of the 2011 Great East Japan Earthquake, satellites with very high resolution from the International Charter were used for availability monitoring.

Numerous interviewees (users) said they would find many types of information useful to assess damage after a mass disaster and would appreciate when satellite images and aerial photos are supplied quickly at no cost. Aerial photos provide very-high-resolution images of small areas and are complementary to satellite images.

In future, data providers should continue to improve the emergency observation service to meet user needs, such as data type (optical or SAR, high resolution or wide range), response time, value adding, and reliability and validation of value-added products, under strong

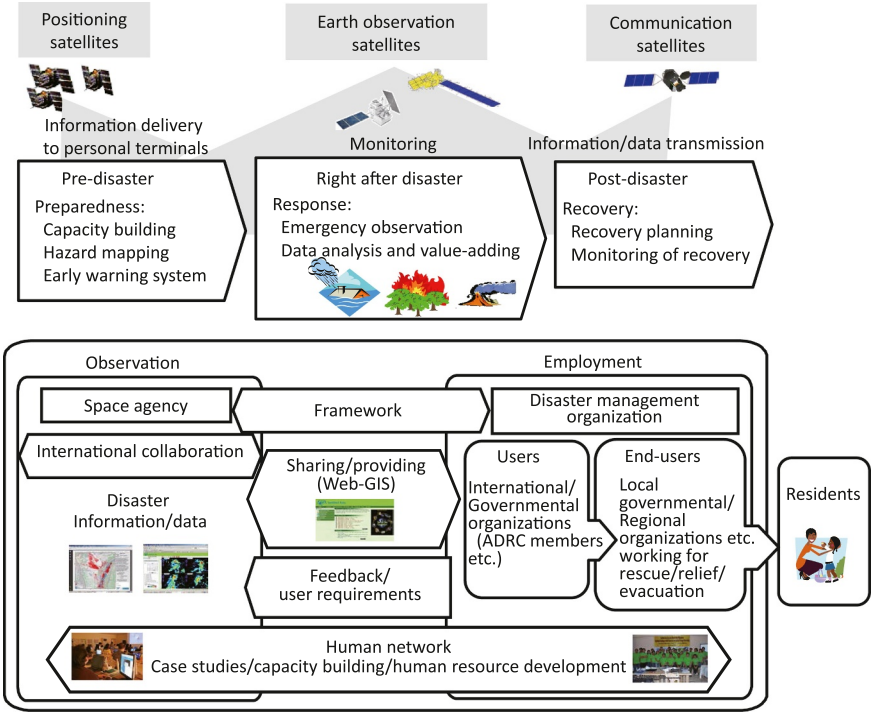


Fig. 14. Conceptual illustration of applying satellite remote sensing to disaster management support.

management and coordination over all the processes of emergency observation, including feedback from users.

(5) Sharing/providing system of disaster information/data
A sharing/providing system of data/information is essential for conducting disaster management support. It should function to share disaster information and data on Web-GIS, and to transmit data to authorized persons. It also has a secure-access control function to limit users from viewing and downloading data according to the data policies of the data providers. Communication satellites and positioning satellites are useful for satellite data transmission, evacuation warning message delivery, and others (for example, refer to Section 4.2.3).

(6) International collaboration
International collaboration is essential for disaster management support, particularly in the case of catastrophic disasters such as the 2011 Great East Japan Earthquake [12]. As many satellites as possible are expected to support disaster management, particularly response, to maximize results based on: (i) the frequency of observation; (ii) rapid response, that is, to provide satellite-based disaster information to users (disaster responders) as quickly as possible (refer to Section 4.2.1); (iii) wide area coverage; (iv) types of satellite data with respect to high-resolution or wide-ranging surveillance, as well as SAR (X- or C- or L-band) or optical sensors; and (v) the uncertainty of getting excellent imagery depending on weather conditions. Some international space-based initiatives are contributing to the field of disaster management. Such initiatives include SA, the International Charter, the International Working Group on Satellite-based Emergency Mapping (IWG-SEM) by a voluntary group of organizations involved in satellite-based emergency mapping, the Committee on Earth-observation Satellites (CEOS), Copernicus (a European Union program), UN-SPIDER (a United Nations program), and the Global Earth-observation System of Systems (GEOSS) of GEO.

(7) Feedback from users
Feedback from users on each disaster event is valuable to data providers, whether it goes well or not. If not successful, user feedback contains lessons learned for improvement of activities; if successful, it is a measure and evidence for the results of data providers. However, there are difficulties in collecting feedback from users, as mentioned in Section 4.3. One solution is site surveys after disaster response support visiting users and end-users who worked for disaster response (refer to Section 3.1), to confirm in detail how satellite-based disaster data/information were employed and to discuss issues related to provided support.

Sentinel Asia (SA) has its origins in the Asia-Pacific region, and its philosophy and strategy (that is, its inclusiveness, highly collaborative nature, staged implementation strategy, case studies and holistic approach), could be extended to other regions, with reference to the above-derived seven requirements for implementation.

Finally, users request continuity of activities; data providers (space agencies and others) need sustained efforts. There is a saying, “Disasters befall us when we least expect them,” by Torahiko Terada (1878–1935), a Japanese scholar of disaster prevention as well as a physicist.

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